

Basic React Router Application

1. Package.json (Dependencies)

json

```
{
  "name": "react-router-app",
  "version": "1.0.0",
  "private": true,
  "dependencies": {
    "react": "^18.2.0",
    "react-dom": "^18.2.0",
    "react-router-dom": "^6.8.0",
    "react-scripts": "5.0.1"
  }
}
```

2. App.js (Main Application)

jsx

```
import React from 'react';
import { BrowserRouter, Routes, Route, Link } from 'react-router-dom';
import './App.css';

// Component Imports
import Home from './components/Home';
import About from './components/About';
import Services from './components/Services';
import Contact from './components/Contact';

function App() {
  return (
    <BrowserRouter>
      <div className="App">
        {/* Navigation Bar */}
        <nav className="navbar">
          <div className="nav-brand">
            <h2>React Router App</h2>
          </div>
          <ul className="nav-menu">
            <li>
              <Link to="/" className="nav-link">Home</Link>
            </li>
            <li>
              <Link to="/about" className="nav-link">About</Link>
            </li>
            <li>
              <Link to="/services" className="nav-link">Services</Link>
            </li>
            <li>
              <Link to="/contact" className="nav-link">Contact</Link>
            </li>
          </ul>
        </nav>

        {/* Main Content Area with Routes */}
        <div className="container">
          <Routes>
            <Route path="/" element={<Home />} />
            <Route path="/about" element={<About />} />
            <Route path="/services" element={<Services />} />
            <Route path="/contact" element={<Contact />} />
          </Routes>
        </div>
      </div>
    </BrowserRouter>
  );
}
```

```
export default App;
```

3. App.css (Styling)

CSS

```
* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
}

body {
  font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
  background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
  min-height: 100vh;
}

.App {
  min-height: 100vh;
}

/* Navbar Styles */
.navbar {
  background: white;
  padding: 1rem 2rem;
  display: flex;
  justify-content: space-between;
  align-items: center;
  box-shadow: 0 2px 10px rgba(0,0,0,0.1);
}

.nav-brand h2 {
  color: #667eea;
  font-size: 1.5rem;
}

.nav-menu {
  display: flex;
  list-style: none;
  gap: 2rem;
}

.nav-link {
  text-decoration: none;
  color: #333;
  font-weight: 500;
  padding: 0.5rem 1rem;
  border-radius: 4px;
  transition: all 0.3s ease;
}

.nav-link:hover {
  background: #667eea;
  color: white;
}

/* Container Styles */
.container {
  max-width: 1200px;
  margin: 2rem auto;
  padding: 0 1rem;
}

/* Page Styles */
.page {
  background: white;
  border-radius: 10px;
  padding: 2rem;
  box-shadow: 0 5px 20px rgba(0,0,0,0.1);
}
```

```
.page h1 {
  color: #667eea;
  margin-bottom: 1.5rem;
  font-size: 2.5rem;
}

.page p {
  color: #666;
  line-height: 1.6;
  margin-bottom: 1rem;
}

/* Feature Cards for Services Page */
.feature-grid {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(250px, 1fr));
  gap: 1.5rem;
  margin-top: 2rem;
}

.feature-card {
  background: #f8f9fa;
  padding: 1.5rem;
  border-radius: 8px;
  text-align: center;
  transition: transform 0.3s ease;
}

.feature-card:hover {
  transform: translateY(-5px);
  box-shadow: 0 5px 15px rgba(102, 126, 234, 0.2);
}

.feature-card h3 {
  color: #333;
  margin-bottom: 1rem;
}

.feature-card p {
  color: #666;
  margin: 0;
}

/* Contact Form Styles */
.contact-form {
  max-width: 600px;
  margin-top: 2rem;
}

.form-group {
  margin-bottom: 1.5rem;
}

.form-group label {
  display: block;
  margin-bottom: 0.5rem;
  color: #333;
  font-weight: 500;
}

.form-group input,
.form-group textarea {
  width: 100%;
  padding: 0.75rem;
  border: 1px solid #ddd;
  border-radius: 4px;
  font-size: 1rem;
  transition: border-color 0.3s ease;
}

.form-group input:focus,
.form-group textarea:focus {
  outline: none;
  border-color: #667eea;
}
```

```

}

.form-group textarea {
  height: 150px;
  resize: vertical;
}

.submit-btn {
  background: #667eea;
  color: white;
  border: none;
  padding: 1rem 2rem;
  font-size: 1rem;
  border-radius: 4px;
  cursor: pointer;
  transition: background 0.3s ease;
}

.submit-btn:hover {
  background: #764ba2;
}

/* Team Section for About Page */
.team-section {
  margin-top: 2rem;
}

.team-grid {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(200px, 1fr));
  gap: 1.5rem;
  margin-top: 1.5rem;
}

.team-member {
  text-align: center;
  padding: 1rem;
  background: #f8f9fa;
  border-radius: 8px;
}

.member-icon {
  font-size: 3rem;
  margin-bottom: 1rem;
}

.team-member h3 {
  color: #333;
  margin-bottom: 0.5rem;
}

.team-member p {
  color: #666;
  margin: 0;
}

/* Responsive Design */
@media (max-width: 768px) {
  .navbar {
    flex-direction: column;
    padding: 1rem;
  }

  .nav-brand {
    margin-bottom: 1rem;
  }

  .nav-menu {
    gap: 1rem;
    flex-wrap: wrap;
    justify-content: center;
  }

  .page h1 {

```

```
    font-size: 2rem;
  }
}
```

4. components/Home.js

jsx

```
import React from 'react';

function Home() {
  return (
    <div className="page">
      <h1>Welcome to Our Website</h1>
      <p>
        This is the home page of our React Router demonstration application.
        We're using React Router v6 to handle navigation between different pages
        without page refresh, providing a smooth single-page application experience.
      </p>
      <p>
        Feel free to explore our website using the navigation menu above.
        You can learn more about us, see our services, or get in touch through
        the contact page.
      </p>

      <div style={{ marginTop: '2rem', padding: '1rem', background: '#f8f9fa', borderRadius: '8px' }}>
        <h3 style={{ color: '#667eea', marginBottom: '1rem' }}>Quick Features:</h3>
        <ul style={{ listStyle: 'none', padding: 0 }}>
          <li style={{ marginBottom: '0.5rem' }}>✔ Client-side routing with React Router</li>
          <li style={{ marginBottom: '0.5rem' }}>✔ No page refresh on navigation</li>
          <li style={{ marginBottom: '0.5rem' }}>✔ Responsive navigation bar</li>
          <li style={{ marginBottom: '0.5rem' }}>✔ Beautiful gradient background</li>
          <li style={{ marginBottom: '0.5rem' }}>✔ Modern UI design</li>
        </ul>
      </div>
    </div>
  );
}

export default Home;
```

5. components/About.js

jsx

```
import React from 'react';

function About() {
  return (
    <div className="page">
      <h1>About Us</h1>
      <p>
        We are a passionate team dedicated to creating amazing web experiences
        using modern technologies like React and React Router. Our mission is
        to build fast, responsive, and user-friendly applications that make a
        difference.
      </p>

      <div className="team-section">
        <h2 style={{ color: '#667eea', marginBottom: '1rem' }}>Our Team</h2>
        <div className="team-grid">
          <div className="team-member">
            <div className="member-icon">      </div>
            <h3>John Doe</h3>
            <p>Lead Developer</p>
          </div>

          <div className="team-member">
            <div className="member-icon">      </div>
            <h3>Jane Smith</h3>
            <p>UI/UX Designer</p>
          </div>
        </div>
      </div>
    </div>
  );
}
```

```

    </div>

    <div className="team-member">
      <div className="member-icon">□ </div>
      <h3>Mike Johnson</h3>
      <p>Backend Developer</p>
    </div>

    <div className="team-member">
      <div className="member-icon"> </div>
      <h3>Sarah Williams</h3>
      <p>Project Manager</p>
    </div>
  </div>
</div>
</div>
</div>
);
}

```

```
export default About;
```

6. components/Services.js

jsx

```

import React from 'react';

function Services() {
  const services = [
    {
      title: 'Web Development',
      description: 'Custom web applications built with React, Node.js, and modern technologies.'
    },
    {
      title: 'Mobile Development',
      description: 'Cross-platform mobile apps using React Native for iOS and Android.'
    },
    {
      title: 'UI/UX Design',
      description: 'Beautiful and intuitive user interfaces designed with user experience in mind.'
    },
    {
      title: 'Cloud Solutions',
      description: 'Scalable cloud infrastructure and deployment solutions for your applications.'
    }
  ];

  return (
    <div className="page">
      <h1>Our Services</h1>
      <p>
        We offer a comprehensive range of services to help you build and scale
        your digital products. From initial concept to final deployment, we've
        got you covered.
      </p>

      <div className="feature-grid">
        {services.map((service, index) => (
          <div key={index} className="feature-card">
            <h3>{service.title}</h3>
            <p>{service.description}</p>
          </div>
        ))}
      </div>
    </div>
  );
}

export default Services;

```

7. components/Contact.js

jsx

```
import React, { useState } from 'react';

function Contact() {
  const [formData, setFormData] = useState({
    name: '',
    email: '',
    message: ''
  });

  const handleChange = (e) => {
    setFormData({
      ...formData,
      [e.target.name]: e.target.value
    });
  };

  const handleSubmit = (e) => {
    e.preventDefault();
    alert('Thank you for your message! We will get back to you soon.');
```

Fill out the form below and we'll get back to you as soon as possible.

```
    setFormData({ name: '', email: '', message: '' });
  };

  return (
    <div className="page">
      <h1>Contact Us</h1>
      <p>
        Have a question or want to work together? We'd love to hear from you!
        Fill out the form below and we'll get back to you as soon as possible.
      </p>

      <div style={{ display: 'grid', gridTemplateColumns: '1fr 1fr', gap: '2rem', marginTop: '2rem' }}>
        <div>
          <h3 style={{ color: '#667eea', marginBottom: '1rem' }}>Get in Touch</h3>
          <p><strong> Address:</strong> 123 Tech Street, Silicon Valley, CA 94025</p>
          <p><strong> Phone:</strong> (555) 123-4567</p>
          <p><strong>✉ Email:</strong> info@reactrouterapp.com</p>
          <p><strong> Hours:</strong> Mon-Fri 9am - 6pm PST</p>
        </div>

        <form className="contact-form" onSubmit={handleSubmit}>
          <div className="form-group">
            <label htmlFor="name">Name:</label>
            <input
              type="text"
              id="name"
              name="name"
              value={formData.name}
              onChange={handleChange}
              required
            />
          </div>

          <div className="form-group">
            <label htmlFor="email">Email:</label>
            <input
              type="email"
              id="email"
              name="email"
              value={formData.email}
              onChange={handleChange}
              required
            />
          </div>

          <div className="form-group">
            <label htmlFor="message">Message:</label>
            <textarea
              id="message"
              name="message"
              value={formData.message}
              onChange={handleChange}
            />
          </div>
        </form>
      </div>
    </div>
  );
}
```

```
        required
      ></textarea>
    </div>

    <button type="submit" className="submit-btn">Send Message</button>
  </form>
</div>
</div>
);
}

export default Contact;
```

8. index.js

```
jsx

import React from 'react';
import ReactDOM from 'react-dom/client';
import App from './App';

const root = ReactDOM.createRoot(document.getElementById('root'));
root.render(
  <React.StrictMode>
    <App />
  </React.StrictMode>
);
```

Output Screenshots (Text Description)

Home Page Output:

```
text

-----
|                               NAVBAR                               |
|  React Router App  | Home | About | Services | Contact |
-----

WELCOME TO OUR WEBSITE
=====
This is the home page of our React Router demonstration
application. We're using React Router v6 to handle navigation
between different pages without page refresh...

Quick Features:
• Client-side routing with React Router
• No page refresh on navigation
• Responsive navigation bar
• Beautiful gradient background
• Modern UI design
```

About Page Output:

```
text

-----
|                               NAVBAR                               |
|  React Router App  | Home | About | Services | Contact |
-----

ABOUT US
=====
We are a passionate team dedicated to creating amazing web
experiences using modern technologies like React and React Router...

OUR TEAM
-----
      John Doe          Jane Smith      □      Mike Johnson
Lead Developer    UI/UX Designer    Backend Developer
```


Services Page Output:

text

```
-----
|                               NAVBAR                               |
|  React Router App  | Home | About | Services | Contact |
|-----|
```

OUR SERVICES

We offer a comprehensive range of services to help you build and scale your digital products...

WEB DEVELOPMENT
Custom web apps
with React, Node.js

MOBILE DEVELOPMENT
Cross-platform apps
using React Native

UI/UX DESIGN
Beautiful and
intuitive interfaces

CLOUD SOLUTIONS
Scalable cloud
infrastructure

Contact Page Output:

text

```
-----
|                               NAVBAR                               |
|  React Router App  | Home | About | Services | Contact |
|-----|
```

CONTACT US

Have a question or want to work together?...

GET IN TOUCH

123 Tech Street
Silicon Valley

(555) 123-4567

✉ info@company.com

Mon-Fri 9am-6pm

CONTACT FORM

Name: []

Email:[]

Message:
[]
[]

[Send Message]

Key Features Implemented:

- React Router Setup:** Using BrowserRouter, Routes, and Route components
- Navigation:** Link components for client-side routing
- Responsive Design:** Mobile-friendly navbar with CSS
- Modern UI:** Gradient background, cards, and smooth animations
- Form Handling:** Contact form with state management

6. **No Page Refresh:** Smooth navigation between pages

How to Run:

1. Create a new React app: `npx create-react-app react-router-app`
2. Install React Router: `npm install react-router-dom`
3. Replace the default files with the code above
4. Run: `npm start`
5. Open <http://localhost:3000>

The application will have a beautiful purple gradient background with a white navigation bar and content cards, providing a professional and modern user interface with seamless page transitions.